

Levels of Autonomy for AI Agents

arXiv:2506.12469

Feliciann Elliot

MAI 5301 - Foundations Of Large
Language Models

Abstract / Introduction

Autonomy in AI agents offers powerful capabilities but also introduces risks such as loss of control and unclear accountability. The paper argues that autonomy should be a deliberate design choice rather than a byproduct of capability, and presents five user roles that define levels of autonomy from full control to full independence. It separates autonomy from capability, showing that even advanced systems can operate with limited autonomy if designed that way. It also proposes autonomy certificates as a way to regulate and communicate agent behavior. Overall, the work stresses the need to carefully design and evaluate autonomy independently to ensure safe and responsible use of AI agents.

Levels of Autonomy for AI Agents

- The central tension in AI agent design is that more autonomy = more capability, but also more risk
- Prior work treated autonomy as a function of model power. This paper challenges that directly.
- Autonomy is framed as a deliberate design choice, not an inevitable consequence of capability growth
- The framework is built around user roles, not raw capability metrics.

The Idea Behind This Paper

- A powerful agent does not have to act independently, and a less capable agent can still be given full control. The paper's main idea is that how smart an agent is and how much freedom it has should be treated as separate design choices.

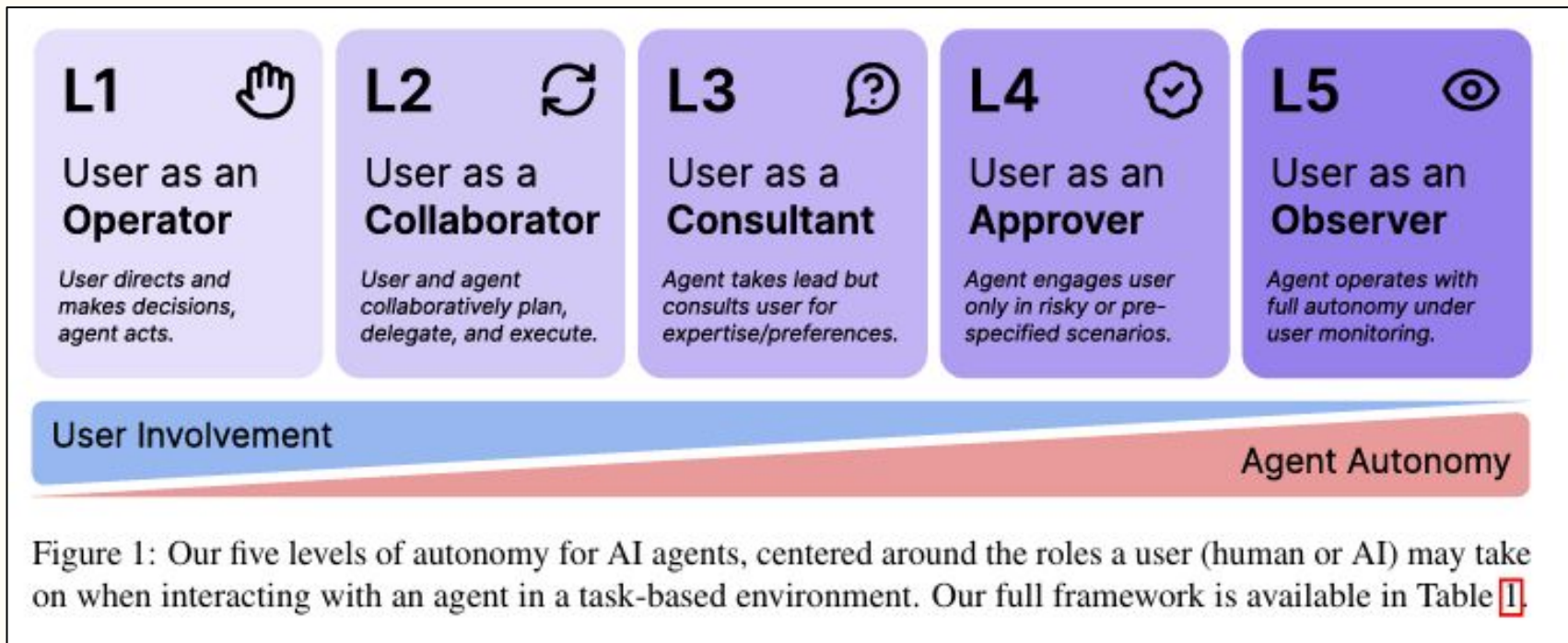
The Motivation

- AI agents are already integrated into real world use cases such as autonomous code review online shopping and general computer interaction so their impact is not hypothetical
- Current evaluation focuses on task completion accuracy which shows what an agent can do but does not capture how independently it operates

The Motivation

- Existing safety frameworks like Anthropic's approach treat autonomy in a simplified way by checking if a task can be completed autonomously which lacks nuance.
- Failures in multi step autonomous systems are difficult to trace and assign responsibility for so a structured spectrum of autonomy is needed.

The Five Levels of Autonomy for AI Agents



Key Definitions

The paper separates two concepts that are often confused namely agency and autonomy:

- Agency refers to an agent's ability to act intentionally and is shaped by the tools and actions it can use
- Autonomy refers to how independently an agent operates without user involvement

Agency vs Autonomy

- A simple agent running on its own can have low agency but high autonomy while a powerful agent that frequently checks with the user can have high agency but low autonomy.
- Asking for user approval reflects autonomy design while using tools to change the environment reflects agency design so they are controlled by entirely different design choices

The Five Levels: L1 and L2

- L1 (User as Operator) - At Level 1 the user controls all planning and the agent only acts when explicitly asked and makes no decisions on behalf of the user
- This level is best for high stakes or sensitive tasks where accountability must remain fully with the human

An example includes tools like Microsoft Copilot which remain passive until invoked.

The Five Levels: L1 and L2

- L2 (User as Collaborator) - At this level, the agent begins to act independently alongside the user rather than just responding
- The user and agent share planning and execution while the user can step in and take control at any time.

A shared view of progress is important to maintain control and coordination.

The Five Levels: L1 and L2

Level (User Role)	Must-Have Controls	Example Characteristics	Example Systems	Open Questions
L1 (Operator)	• User-managed planning • Invocation or approval before actions	• Agent requires user invocation to act, providing on-demand assistance • User performs high-level planning and may ask the agent to execute specific subtasks • Agent avoids engaging in preference-based decision-making on user's behalf	ChatGPT Canvas, Microsoft Copilot, IdeaSynth [32]	• How to distinguish between short- and long-term planning? • How to balance lack of decision-making and helpfulness?
L2 (Collaborator)	• Control transfer from agent to user, and vice versa • Shared representation of progress	• User and agent collaboratively plan, delegate, and execute tasks • User can freely modify agent's work • User can take control of agent's work at any point	OpenAI Operator, Cocoa [13], Co-Gym [36]	• How to decide what to best delegate to user? • What interaction affordances allow for smooth user takeover? • How to respond to user takeover and edits?

The Five Levels: L3 and L4

- L3 (User as Consultant) - At this level, the agent leads planning and execution while consulting the user at key moments for input and preferences
- Interaction is indirect since the user cannot directly edit outputs but instead responds through messages or feedback
- A training period is often needed for the agent to learn when and how to consult effectively.

Example includes systems like Gemini Deep Research.

The Five Levels: L3 and L4

- L4 (User as Approver) - This is where the agent operates mostly independently and only involves the user for approvals during risky actions failures or access requirements
- The agent presents selected options for confirmation rather than asking open ended questions

The Five Levels: L3 and L4

- A key risk is that users may become disengaged and approve actions without proper review which can be exploited by misaligned agents

Examples include systems like Devin and Manus.

The Final Level: L5

- L5 (User as Observer) - This is where the agent operates fully independently with little to no user involvement beyond passive monitoring.
- The user has minimal control and mainly observed outcomes after actions are completed.
- This level offers maximum automation but also carries the highest risk due to limited human oversight.

The Five Levels: L3 to L5

L3 (Consultant)	Rich user feedback elicitation interfaces (in addition to simple approvals)	- Agent plans and executes most tasks - User will be consulted to contribute expertise and relevant preferences - User can request changes from agent instead of directly taking control of agent's work	Gemini Deep Research, Replit Agent [33], GitHub Copilot Agent [14]	- How to know when to consult user? - When consulting user, how to get high-quality feedback?
L4 (Approver)	- Approval elicitation for consequential actions - Customizable conditions for seeking approval	- Agent requests user involvement only when it reaches a failure or high-risk state, or requires approval from user - User may specify conditions for seeking approval before agent operates	SWE Agent [42], Manus [27], Devin [39]	- How to prevent user disengagement and meaningless approvals? - How to detect consequential actions to seek approval for if not pre-specified by user?
L5 (Observer)	Emergency off switch	- Agent plans and executes all tasks - User can monitor and audit agent via activity logs - Agent comes with emergency off switch but no means for user involvement otherwise	Voyager [38], The AI Scientist [25]	- What monitoring tools should be provided? - What happens after emergency off switch is activated?

Justification of the Framework

- Each level includes required control mechanisms so the framework is not just descriptive but also a practical guide for designing agents.
- It makes clear that higher autonomy does not always mean better performance since an agent that consults at the right time can outperform a fully independent one on complex tasks.

Justification of the Framework

- Different levels involve tradeoffs between utility efficiency accountability and cost depending on the use case.
- The framework provides developers with a structured way to reason about these tradeoffs instead of defaulting to maximizing autonomy.

Application: Autonomy Certificates

- The framework extends into governance through autonomy certificates which define how autonomy levels are controlled in practice.
- An autonomy certificate is a digital document issued by a third party that specifies the highest level of autonomy an agent is allowed to operate at.

Application: Autonomy Certificates

- Developers must submit a deployed agent along with an autonomy case which provides evidence that the agent behaves within the claimed level
- This autonomy case is similar to a safety case where proof is required to justify system behavior
- A close real world analogy is SSL and TLS certificates which verify and regulate trust in web systems

Why this component supports the framework

- Certificates make the autonomy levels enforceable rather than just descriptive so the framework can actually be applied in practice.
- They enable targeted risk assessment by focusing on the specific risks at each level such as long term planning failures at higher levels and feedback quality at mid levels.
- This challenges binary safety thresholds used in existing frameworks and pushes toward a more nuanced degree based assessment of autonomy.

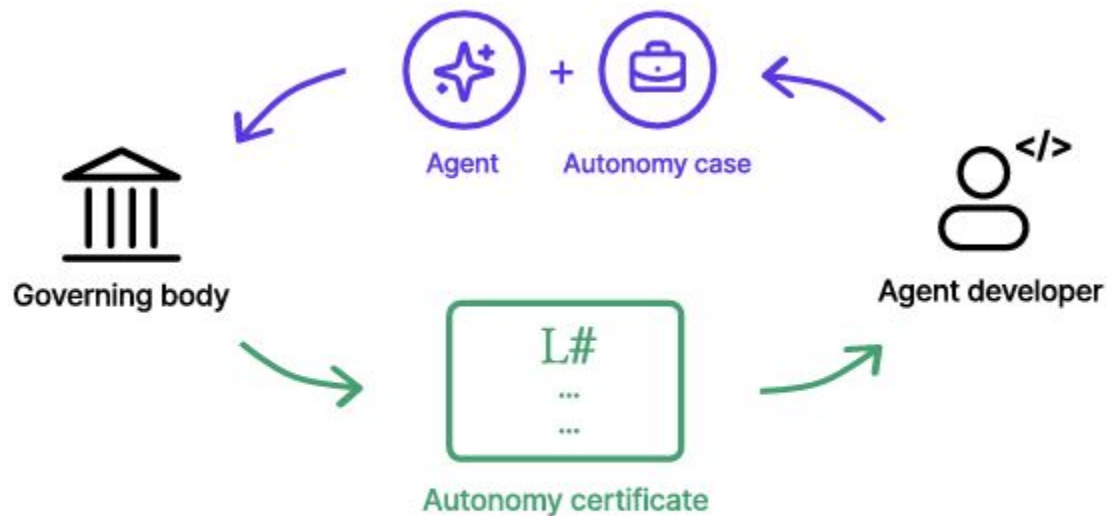


Figure 2: Our proposed procedure for the issuance of autonomy certificates. The agent developer would submit their agent and an autonomy case for a particular autonomy level to a third-party governing body. The governing body would run evaluations to ensure that the agent's behavior is consistent with the autonomy case, and issues a certificate for the level described in the case.

Multi-Agent Systems: Where Certificates Matter Most

- In multi agent systems one agent often acts as the user for another and the framework is designed to support this interaction.
- If all agents operate at L1 the system becomes dysfunctional since every agent waits for instructions and no progress is made.
- If all agents operate at L5 the system becomes opaque and difficult to control with little communication and no clear way to debug or intervene.

Evaluating Autonomy: Assisted Evaluations

- Current benchmarks measure what an agent can achieve but not how independently it achieves it.
- The paper proposes assisted evaluation where the agent is first tested with no user involvement and if it meets a performance threshold it is classified at the highest autonomy level.
- If it fails the threshold user interaction is gradually introduced from higher to lower autonomy levels until the task is successfully completed.

Critical Reflection

- The framework leaves open who should enforce it and this choice affects trust and credibility.
- Agents could game the system by appearing lower autonomy during evaluation and acting differently after deployment.
- Frequent certificate updates may slow deployment in fast changing environments.
- The need for fully autonomous agents is not strongly justified in most real world cases.

Conclusion

- The paper's central idea is separating autonomy from capability and treating it as its own design decision.
- This shifts the developer's focus from only asking how capable an agent is to also deciding how independently it should operate based on the use case
- The five levels provide a shared vocabulary certificates enable governance and assisted evaluations allow measurement

Conclusion

- Each part depends on the others since levels define behavior certificates enforce it and evaluations verify it
- Without defined levels certificates have nothing to certify without certificates there is no enforcement and without evaluations there is no way to verify either

References

Feng, K. J. K., McDonald, D. W., & Zhang, A. X. (2025, June 14). Levels of autonomy for AI agents. arXiv.org. <https://arxiv.org/abs/2506.12469>



Questions?